

INMAS: Internship Network in the Mathematical Sciences

YEAR 4 EVALUATION REPORT

MARY A MCEATHRON, PHD

Table of Contents

INTRODUCTION	2
EVALUATION ACTIVITIES	2
EVALUATION FINDINGS	3
TECHNICAL AND PROFESSIONAL DEVELOPMENT WORKSHOPS.....	3
FEEDBACK FROM FACULTY AND STAFF	5
<i>Awareness about INMAS.....</i>	<i>5</i>
<i>Departmental Support for Industry Careers.....</i>	<i>6</i>
<i>Additional Impacts of INMAS</i>	<i>8</i>
<i>INMAS's Value and Future Sustainability</i>	<i>9</i>
<i>What Next for Year 5?</i>	<i>10</i>

Introduction

The Internship Network in the Mathematical Sciences (INMAS), a five-year project funded by the National Science Foundation (NSF), is co-directed by Fadil Santosa (Johns Hopkins University) and Richard Laugesen (University of Illinois, Urbana-Champaign). It is currently in its fourth year. The goals of this project are to:

- Provide high quality training and internship experiences that broaden career opportunities for students in the Mathematical Sciences.
- Bring powerful computational and modeling tools to bear on industry's most pressing problems.
- Build mutually beneficial connections between industries and mathematics departments.
- Support systems change within mathematics departments in support of expanded opportunities for students and industry partners.

The major activities planned to accomplish these goals are the following.

- Launch regional networks, with hubs at Johns Hopkins University and the University of Illinois and nodes at graduate level institutions in neighboring states.
- Prepare students for the internships through technical and professional skills training.
- Seek out challenges in industry and government organizations that can be tackled by Mathematical Sciences techniques.
- Create internship projects at those host organizations, and match with students in the network.

Evaluation Activities

Evaluation activities over the last four years included surveying students about their experiences in the technical workshops, surveying students to determine growth in technical and professional skills, interviewing students about their internship experiences, and interviewing company representatives about their experiences hosting a student intern. For the fourth year, we continued the process of surveying student participants and added the task of conducting interviews with faculty and staff at participating universities.

To review, the evaluation activities for the fourth year included the following:

- At the beginning of the program, an online Pre-Training Survey was emailed to 67 students in October 2023 to learn about their technical and professional skill levels prior to any INMAS workshops as well as perceptions about industry opportunities. Fifty-six students completed the pre-training survey.
- Following each weekend technical workshop, a feedback and process survey link was emailed to students as follows:¹
 - October 2023: Python Programming (52 respondents)
 - November 2023: Statistical Methods using Python (49 respondents)
 - January 2024: Machine Learning (34 respondents)

¹ The results of these feedback surveys were shared with INMAS staff directly after each workshop to support continuous improvement. This report will only present the pre-post survey results.

- April 2024: Post-Training Survey (41 respondents)
- Interviews were conducted in Summer 2024 with faculty and staff of participating university departments. Thirteen people from nine departments participated in the interviews.

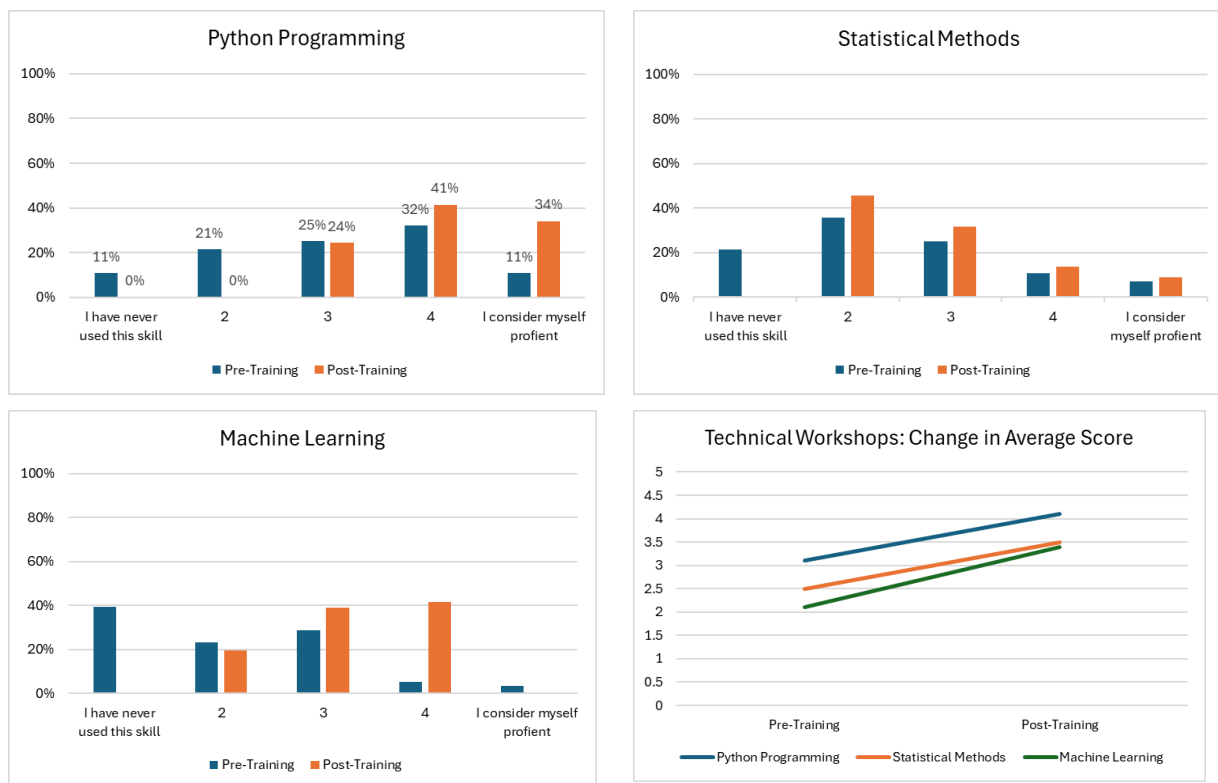
Evaluation Findings

This report includes the results of the pre- and post-training surveys given to students to learn about changes in their technical and professional skills. It also includes the findings from interviews conducted with faculty and staff. The interviews were conducted via Zoom and were recorded with participants' permission. Interview recordings were transcribed and uploaded into NVivo, a program that supports qualitative analysis. Direct quotations from the interviews are used to illustrate thematic points.

Technical and Professional Development Workshops

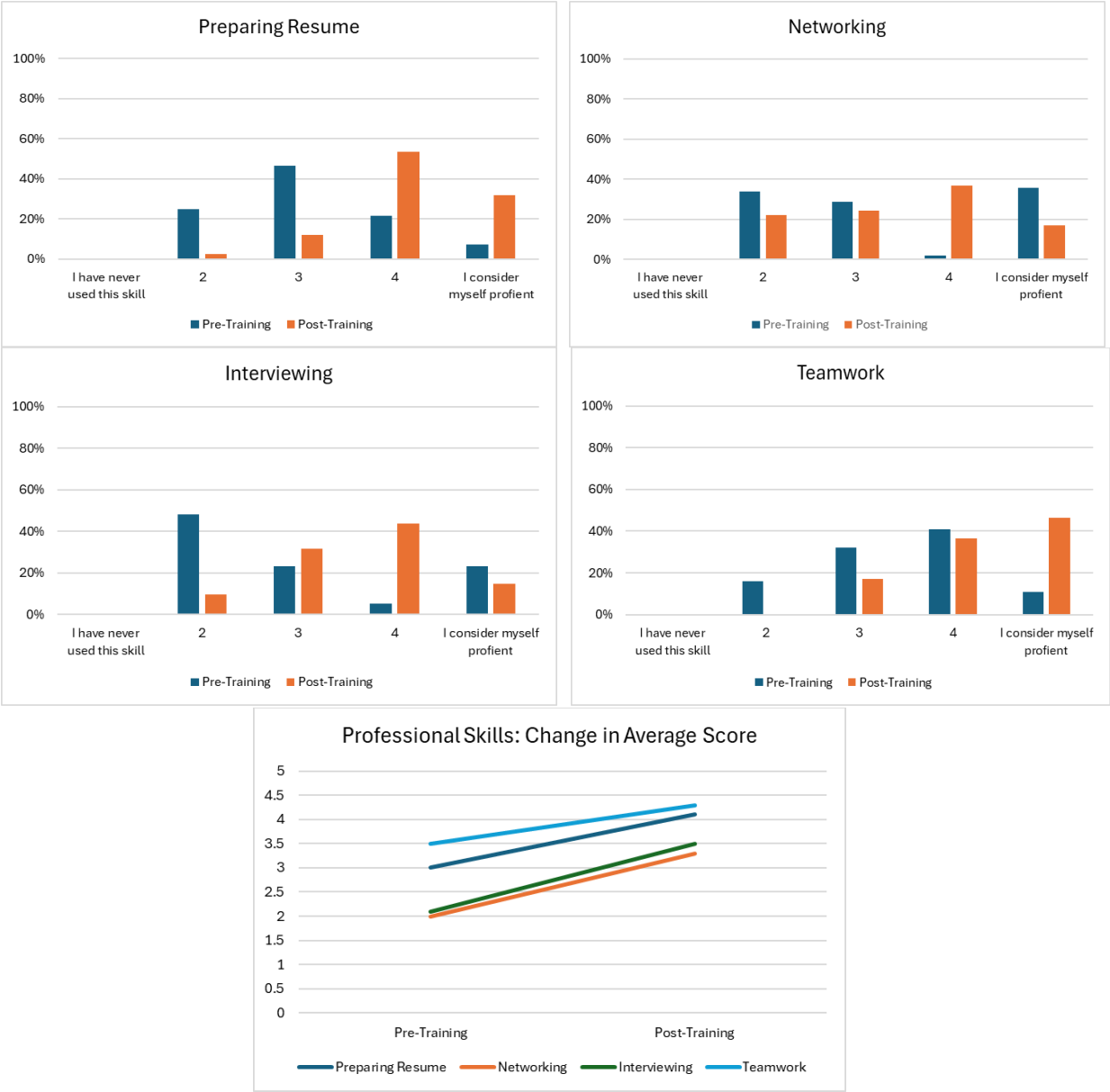
The following are data from the pre- and post-workshop surveys. The pre-workshop survey was emailed in October 2023 prior to any trainings; the post-workshop survey was emailed in April 2024. Students were asked to rate their level of knowledge on a 1-5 scale: I have never used this skill (1) to I am proficient (5).

Figure 1. Pre- and Post-Training Survey Results: How would you rate your proficiency in the following technical skills?



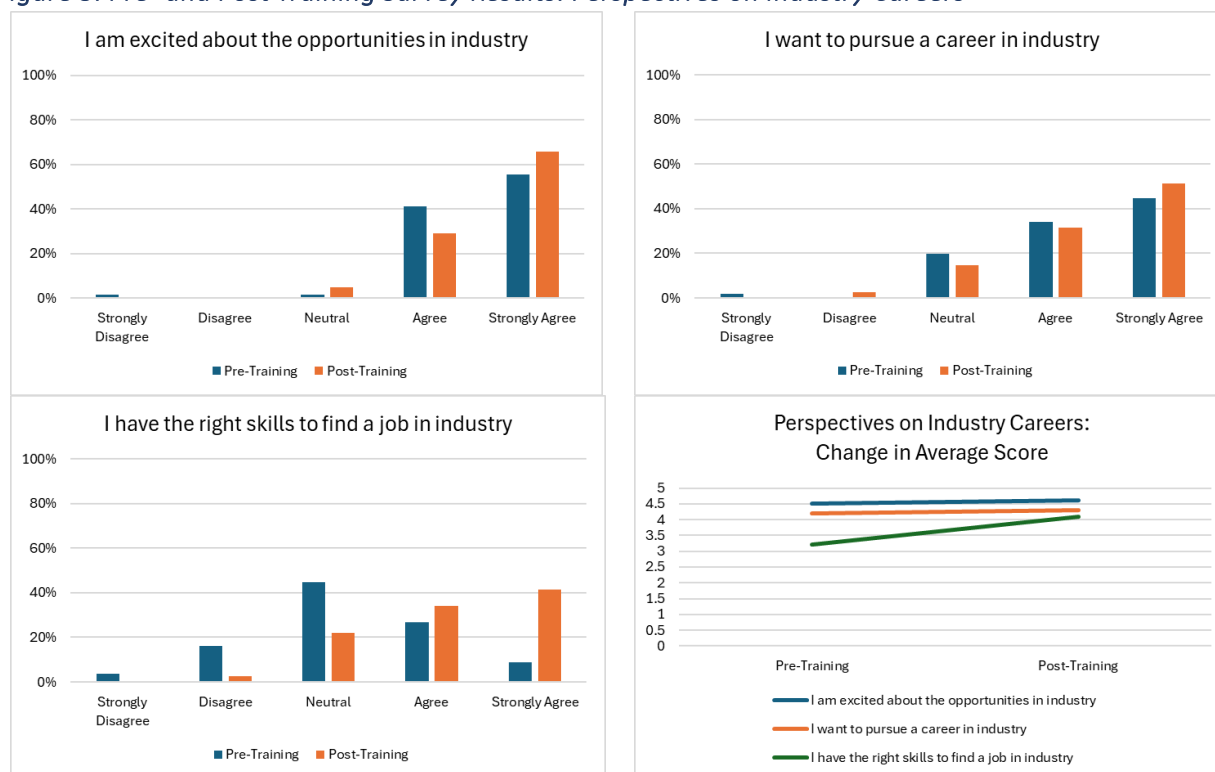
Professional development activities focused on resume and interviewing preparation along with teamwork (via group projects). The following are the survey results for these areas.

Figure 2. Pre- and Post-Training Survey Results: How would you rate your proficiency in the following professional skills?



Students' perceptions of industry careers were assessed at the beginning of the program and at the end of the technical and professional development workshops via the pre- and post-workshop surveys.

Figure 3. Pre- and Post-Training Survey Results: Perspectives on Industry Careers



Feedback from Faculty and Staff

One thread that weaves through all of INMAS is the range of differences among all the levels of participation. Students varied in their technical and professional skills and in their areas of mathematical knowledge and research. Companies varied in size, focus of internship, problems to be solved, and experience in hosting interns. Departments and faculty varied in their interest and experience in engaging with industry, the structure of the department, and their level of involvement with INMAS. It is remarkable, then, that the majority of participants over the last four years—students, industry representatives, and faculty—all report highly positive responses to INMAS.

The faculty and staff who participated in the interviews this year included those who identified as theoretical mathematicians with little or no experience with industry; others were applied mathematicians with deep connections to labs or companies in their geographic area. Some knew the co-directors or the program from the beginning of the program and others were new to both the program and their department positions. Their roles included faculty, directors of graduate studies, department chairs, and INMAS liaisons. In the following presentation of the themes that emerged from the report, I note when this variation led to a differing response.

Awareness about INMAS

Participants shared that there was a range of awareness about INMAS within their departments.

Approximately 20% of the interviewees said that there was full awareness of INMAS throughout the entire department; 30% thought that at least half of the faculty were aware. A few people reported that they thought the level of awareness was quite low among faculty. In those cases, they listed specific people who knew about INMAS such as the director of graduate studies. Note: these levels of awareness were based on the perceptions of the interviewee.

“The faculty is quite well aware. We discussed it in the faculty meeting because we were supposed to make a payment, a membership fee or whatever, to support the students. I informed the department and everybody was excited about it. Everybody thought that it would be a very good idea.”

“At this point? I think everybody's aware the program exists. I do make sure to include everybody in the mailing list. I do talk to the graduate committee about this. I have spoken with the faculty who is involved in recruitment, and we even bring it up when we are trying to recruit students.”

“With faculty members? Less than I would hope. I think most faculty members are kind of vaguely aware it exists and think, oh, that's probably a good idea. But I don't think they've thought very much about it unless a student asks them, can I do this summer?”

Interviewees also pointed out that students were often a conduit of awareness and information within a department, sometimes even more than faculty.

“I think the students all know about it. They talk to each other. I've recommended it to my graduate students and they say, ‘Oh, this person did it.’ So it's definitely out there among the students. ... Everybody who's heard about it has been enthusiastic, but I really think it's been coming from the students more.”

However, one person noted that some students resisted the idea of participating an internship program.

“I've also been surprised when I've advised my students to look into doing it. There has been quite a lot of pushback from students who feel, ‘Oh, you're trying to channel me. It means you don't have faith in me. You're trying to channel me into a non-academic track.’ I think it's foolish, but it's a common pushback.”

Departmental Support for Industry Careers

Interviewees were asked how open their department is in supporting students who choose industry careers and how INMAS impacted that level of support. Their answers to these two questions unscored the deeper, systemic factors that shape the functioning of these graduate programs. First, the structure of the department influenced the ability and interest of faculty. For those who described the focus of their departments as theoretical or pure math, there was not as much interest or desire from faculty members to support students exploring industry careers. However, it was just that type of department in two cases which also saw a greater role for INMAS. Interviewees also noted that each faculty member acts as their own agent in many situations, which can affect how quickly a departmental culture can change.

“It is harder with the others [faculty members] who do not see any advantage because they work in pure math, like algebra proofs. So they do not see how that can help their student. They're convinced that their student will stay in academia and become an instructor or faculty.

They feel that that's a time wasted instead of student over the summer working on the thesis. ... I'm trying to convince more and more my colleagues to be open to the idea that their students might be interested also in internships outside of academia and that it may affect a little bit of timing to write in their thesis, but it can be beneficial for all, not only for faculty to get the work done but also for their students."

"I think in our department, there are many faculty who think that success means any job that they [students] want to do. Then there are some who are like, well, any job that has something to do with their training. I do know one of the faculty members had a student who went on to be a lecturer. That faculty member was quite explicit that they thought that was not good, not successful."

"I feel like everyone wants what's best for our students. I think on paper, these other things [industry careers] are definitely second choices in the sense that the program [referring to their department] from the ground up is built for academic pure mathematicians. But I don't think the faculty view is that anyone who goes into industry or government or something is second class citizen. ... I think the perceptions are good, and there's also been a trend at our university to think about the PhD in new ways. Even though our program, as I said, is kind of built for pure math from the beginning, if someone applies to the PhD program saying, 'Oh, I'm applying to the PhD program and my ultimate goal is to work for this company,' it would be a little different to us. It would not be a common occurrence, and maybe we wouldn't necessarily know how to handle that because the beginning stages, you kind of want them thinking about pure math and what area they want to do their dissertation in and not what their career goal is. It was never designed as kind of a career kind of degree."

How students are advised is one pragmatic aspect of supporting all career paths. When asked how their departments approached advising students about the different paths after graduate school, many of the interviewees again noted that changes towards a more expansive advising perspective were slowed by academic culture and experiences of current faculty. This sometimes highlighted the importance of a program like INMAS.

"A lot of our faculty, including me, are not particularly prepared to train our students to go to industry. I've been in academia all my life. Most of my colleagues have been in academia all of their professional lives. So we need something outside the department, I believe, and INMAS has been perfect to fill that role for us."

"We've had really great students who go into other things. But INMAS provides this kind of great resource in that you can't just jump from a pure math background into an industry job. So there needs to be something else."

"But from my perspective, I think that [advising] is lacking because we as faculty have been trained to do one thing and be in academia. So we are not really that familiar, or at least most people are not, with all this internship and industry. It would be a great idea to have that awareness. I think that INMAS has done a great job in that regard because at least they have exposed students. ... The fact that they are at least exposed for one or two summers is a great thing, and we kind of depended on INMAS for that exposure."

Additionally, interviewees noted the importance of advising and planning early in the graduate program and to acquire important technical skills.

“I think people tend not to think about industry or teaching focused jobs until their fifth or sixth year and then it's too late for them. I got advice when I was in graduate school that it does not matter how good you are at research, you need a backup plan. That it could be teaching or industry and you need to start thinking about it in your second year of graduate school.”

“Anyway, but I would say that the main one is just either taking some computer science or statistics courses or doing more programming in the research.”

Lastly, interviewees observed that the two tracks—academics and industry—were not mutually exclusive or pre-determined. That, for example, many of their students who had completed theoretical dissertation research found satisfying careers in industry and other moved to industry out of interest after post-doc positions.

“I think our perception, which is maybe a little bit too favorable on our side as sort of academic mathematicians, is that in most cases our students have already a good set of skills and training, even if it's what one may call in pure mathematics that allows them to acquire some methods and techniques in applied mathematics relatively quickly. And to some extent, this is certainly justified by our experience of hearing students who are getting jobs in industry, in finance and insurance and many other branches of business and industry. And they don't come with a particular knowledge in exactly that field.”

“My first grad students targeted research and ended up in postdocs and only pivoted to industry after. They both wanted to pivot to industry for different reasons, and I don't think in either case necessarily out of necessity, but they actually wanted to. But then my last grad student actually wanted to go to a hedge fund. They were interested in really high-level quant positions, top quant positions. So he was very targeted toward that. Even though he did his PhD in a very theoretical analytic probability area, it was still designed to have some connection to the kind of tools that he would be using in the finance world.”

Additional Impacts of INMAS

As noted above, INMAS provided invaluable internship experience and exposure to industry careers along with technical skill workshops in programming, statistics, and machine learning, which helped provide a bridge to working in more applied settings. Interviewees were asked how INMAS affected their department's relationships with industry. Most replied that they, and other faculty, often relied on INMAS for those connections primarily due to the constraints of time. One department, which already had strong connections with industries and labs, experienced a growth of connections with smaller companies due to the involvement of INMAS.

“Well, with respect to INMAS, they have an internship coordinator who's making contact with the companies. I'm not interacting with the companies, and I don't think anyone from here is interacting with the company. So that's what I mean when I say we're one step removed. To your second question, am I interested? Yeah, I mean, if I can speak frankly, I'm interested in doing a lot of things and I don't have the time. I mean, yeah, it'd be great if I could spend some regular time networking and making contact with industry. I don't know where I would fit that into my

workload.”

Some interviewees also noted that the INMAS experience had a big impact on students, and that actually had a ripple effect within the department.

“I think for students it's very positive. Students are aware that this is a path and this is a stepping stone on that path. Are faculty fully benefiting or fully involved? I don't know for sure, but even without, if not everybody is fully aware in terms of faculty, this knowledge percolates to the students.”

INMAS's Value and Future Sustainability

During the interview I confirmed that Year 5 would be the last year of NSF funding for INMAS. Everyone who participated in the interview, whether they were highly or marginally involved, expressed regret. Interviewees spoke at length about the need for a program like INMAS.

“I think there's a huge need for INMAS, and I don't think it's because the world has shifted. I think there was a need for INMAS 40 years ago. It was just totally ignored. ... For 30 some years it's always been the case that the majority of math PhDs did not get academic jobs. It was kind of viewed, well, that's your problem. I think that there's been an enormous need for INMAS for decades. I would love for it to continue.

It's [the INMAS experience for students] beneficial even if you do get an academic career but we also need to accept the fact that most PhDs are not going to get academic careers. And in fact, I think there's been a horrible shift in our field. The number of postdocs has increased dramatically, which seems like a good thing until you see the effects of it is now that everybody is expected to do two postdocs before they're competitive for a tenure track position. But the number of tenure track positions hasn't increased at all. In fact, if anything, it's contracted. So instead of discovering when you get your PhD that you don't have an academic job and it's time to look elsewhere, you instead discovered four or five years later.”

Interviewees also discussed the components of INMAS that were valuable to their department and students, and what would be challenging to sustain in the absence of INMAS. One of the most frequently discussed components was the importance of dedicated staff for making connections with companies and labs and for running the technical workshops.

“If it becomes one tenth of the DGS's job, we're all going to do a bad job of it. What can they do [in this next year]? To some extent I'm just... it's sad. There's nothing they can do. This needs to be done as a multi-school program.”

“I don't know how much work, once the grant expires, the various personnel are willing to put into running it. I mean, running a workshop for a weekend, that's certainly a non-trivial commitment of time from whoever's running it.”

“But then if it's [university] specific, it is not big enough for us to actually hire a person for it. It's like at the end of the day, what you would like is a large enough scale that you could have at least half a job out of it and not a full job. Not big enough that it's at least half of someone's job. It's going to be really hard to do a good job of it.”

“Is there a coordinator for internships? I don't think so. Not in the department. We have kind of

a message board where when a faculty member or member of staff is aware of some advertisements, we place information there for students to see perhaps.”

The scale of the program, with the hubs and network, was also viewed as a valuable component of INMAS.

“I think it's the right scale that it gives them the better contacts you're going to be able to have. If we try to do this separately at every university, we're each going to have too few contacts. Having some spread of that around is good. Region is a good size also because, for example, there's industry in the Midwest that we can take advantage of, but also, it's just a nice scale. It's like you're really wasting a lot of energy by trying to do this school by school.”

“Another way was really helped was the fact that by having INMAS [we were part of] a network of universities. When you go to the INMAS page [online] you can see John Hopkins and Drexel. So I think also the names help a little bit. ... Industry, they like to see big names and so it gives us as a department more weight.”

Some departments were thinking about or in conversation with others regarding programs which might fill the need once INMAS ends. Interviewees referenced the American Mathematical Society and the Big 10 chair meeting that was held in May 2024. Some departments/students also had access to other internship programs but interviewees always returned to what they liked about INMAS.

“I would really love to see another regionally focused program emerge that we could really get behind. We would really get behind that, especially if it's run the same way. And even the ones I've seen coming down the pike, which I'm glad for, but INMAS had an even better tailored structure and was very, very reasonable [in cost]. The way it was focused on those internships is just wonderful. If we can communicate that if nothing else, that would be wonderfully helpful I think for our students.”

Over half of the interviewees mentioned that they would be willing to contribute more financially, or at least raise that topic within their departments or university, to support a continuation of a program like INMAS. In general, most interviewees were interested in further conversations with INMAS staff about next steps; although there was also a feeling that their departments might revert back to what they were or were not doing before INMAS.

What Next for Year 5?

Interviewees had a few suggestions for INMAS in the coming year of transition. Many were interested in having access to the technical skills workshop materials.

“First thing is that maybe the training, so that was a good part. Maybe INMAS can make it available on the web or something like that so that at least students can see, maybe INMAS has video recordings or whatever they have.”

“I believe that they should help us with whatever they have, the material and stuff to teach those workshops. We've tried before to train our students about machine learning. There are courses in machine learning. I heard from our students that the workshop was very hands-on and the prep focused on the project so that in the end the students can describe it in an interview, for example. ... So I think the way they organize the workshop, the material they have,

if they can pass it on so we have a copy and we have an idea of how it's organized, that will be helpful.”

Others also hoped that INMAS could pass on their institutional knowledge or perhaps return in another form.

“Be willing to transfer that institutional knowledge to the next partners, whoever they may be, even if it's some of the same people. I would love that. I think it was run great and I love the structure and the specific people who were involved too, I think they brought a lot of energy to it. So they have to get a lot of credit, but I hope they would be willing to be a part of something like that in the future.”